Contract No. FA864923P1125

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CLIN001 – List of Data Quality Objectives and Testing Process

The CLIN001 deliverable is a List of Data Quality Objectives (DQOs) and Testing Process. The acceptance criteria for CLIN001 is that "TPOC confirms DQOs and process will inform AFCEC decision makers".

1.1 Task Objective

The objective of Task 1 is to clarify study objectives with the TPOC and internal team, which informs the subsequent program work of establishing the appropriate type of data to be collected on the program (Task 3) and the test plan (Task 4). During this Phase II program, we are using the USEPA's seven-step process as guidance to clarify study objectives, define the appropriate type of data, and specify acceptable levels of potential decision errors that will be used as the basis for establishing the quality and quantity of data needed to support decisions. Data quality objectives (DQOs) are an integrated set of qualitative and quantitative decision statements that define data quality requirements based on the end use of the data. The seven-step Systematic Planning Process (SPP) is outlined in Figure 1. The Statement of Work, Tasks, Milestones, and Timeline as executed in the AFWERX contract are outlined in Table 1.

The work we have completed in Task 1 includes:

- Executed necessary agreements with DAF and Redox Tech Consultant. Redox Tech will support system build and permitting. We have notified Enthalpy Analytical of the award and have begun discussing sampling requirements and protocols with them and Montrose Environmental. Note, we are still working through the specifics of the subcontract with GA, but expect to have it completed by end of August.
- Held a kick-off meeting to establish methods of communication and conducted a pre-mortem meeting to identify key risks for the program,
- Held a kick-off meeting with the TPOC to discuss expectations,
- Reviewed U.S. EPA's seven-step systematic planning process methodologies with new staff members,
- Completed SPP Step 1 – Problem Statement
- Completed SPP Step 2 – Identify the Decision, and
- Partial completion of Step 3 – Identify the Inputs to the Decision, based on what we currently know.

1.2 Step 1. Problem Statement

This section identifies the issues to be addressed in the Phase II program. The overall problem statement is as follows:

AxNano's feed management system must create value for AF hazardous waste treatment beyond SCWO treatment alone.

This overall problem statement consists of three data quality objectives (DQOs), as follows:

1. Objective 1. Can AxNano's SCWO be used to degrade recalcitrant compounds in hazardous waste that are not susceptible to current treatment methods with controllable SCWO reactor conditions and complete degradation/destruction?
2. Objective 2. Can AxNano's SCWO achieve continuous operation with the identified waste streams, reducing high startup energy requirements and become a more sustainable waste treatment process than current treatment methods?
3. Objective 3. Can AxNano's SCWO be integrated into DAF hazmat treatment?

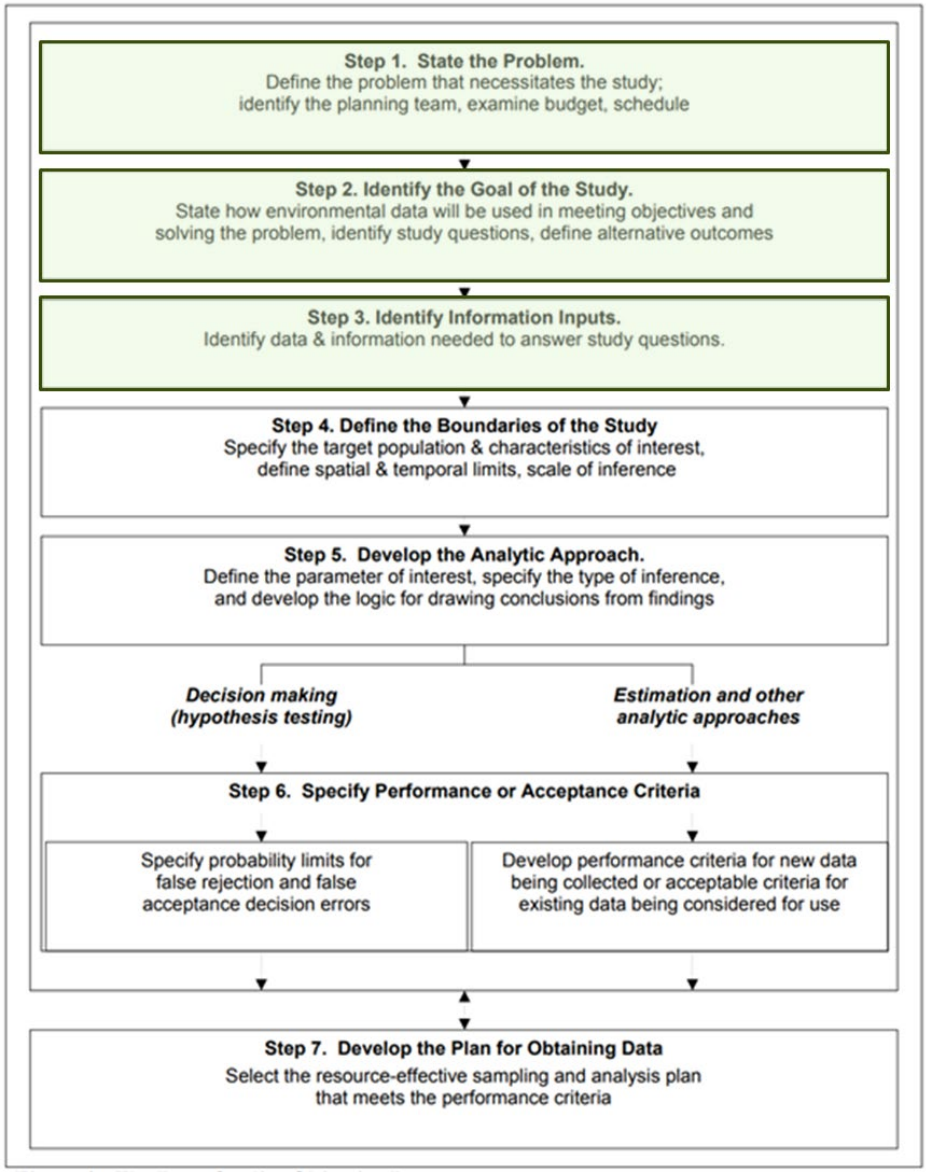

Figure 1. USEPA's seven-step systematic planning process (Task 1 focused on Steps 1-3).

1.3 Step 2. Identify Goals of the Study

The purpose of this step is to define the decision that will be made using data to address the problem. The overall decision to be made based on the data collected under this SBIR Phase II is:

> Can an automated feed management system be designed such that relevant volumes of Air Force-derived wastes can be processed by AxNano's SCWO under continuous operation at a bearable cost and efficacy?

To inform this decision, we have identified three goals of the program that correlate to the above DQOs:

1. Objective 1 Goal: Identify contaminant destruction efficiencies of selected waste streams on the pilot SCWO system and establish feed effects on reactor stability.
2. Objective 2 Goal: Design and test automated waste feed processing equipment with pilot SCWO system.
3. Objective 3 Goal: Ensure AxNano's SCWO (feed management system+SCWO) meets technical (throughput, discharge limits) and economic requirements (cost, logistics) for integration into DAF workflow.

1.4 Step 3. Identify the Inputs to the Decision

Inputs to the Goals listed above are illustrated in Charts 1-3. To achieve high or complete degradation/destruction for targeted hazardous waste species and maintain reactor stability, waste feed rate (information input 1) and waste properties (information inputs 2 to 9) need to be measured and controlled. Some inputs determined in Goal 1 will apply to Goal 2.

1.5 Step 4. Define the Boundaries for the Inputs

Boundary conditions for inputs are listed in Charts 1-3. Boundary conditions are defined as what ranges must the data exist in for the Goals to be achieved. In other words, "what must be true" for the system to work. Some Input's boundaries need to be determined as we secure the wastes for testing (Task 2) and finalize the equipment list (Task 4). For example, COD is closely related with oxidization and will be dependent on the waste composition. Also, destruction efficiency will need to be determined once we determine the exact composition of the waste streams. Some necessary modifications and/or additions of the inputs and boundaries are expected as more facts are learned and fixed in the future. Tasks 1, 2, 3 and 4 of the statement of work (Table 1) will be completed upon achieving the Goal 1.

Once achieving Goal 1, we will need to assess boundary conditions to maintaining the reactor stability and destruction efficiency again with the automated feed processing system toward Goal 2. The feed processing equipment integration may impact reactor stability, therefore while similar Inputs exist for Goal 2 as Goal 1, the boundary conditions may change. The key additional Input toward Goal 2 is design the feed processing equipment with all parts and specifications to be determined after Problem 1 is addressed and more facts are learned and fixed. Tasks 5, 6 and 7 as listed in the Milestone Chart will be completed upon achieving the Goal 2.

Goal 3 of this project is to evaluate how to integrate our system into DAF workflow. Main Inputs to achieve Goal 3 include finding out regulations on air emission and waste discharge at exact treatment locations, feed rate, operating expenses, capital expenses, and other considerations for treatment selection. Customer feedback regarding all the Inputs and Boundaries need to be considered during this evaluation in order to get a reasonable and realistic evaluation technology integration. Tasks 8, 9 and 10 as listed in the Milestone Chart will be completed upon achieving the Goal 3.

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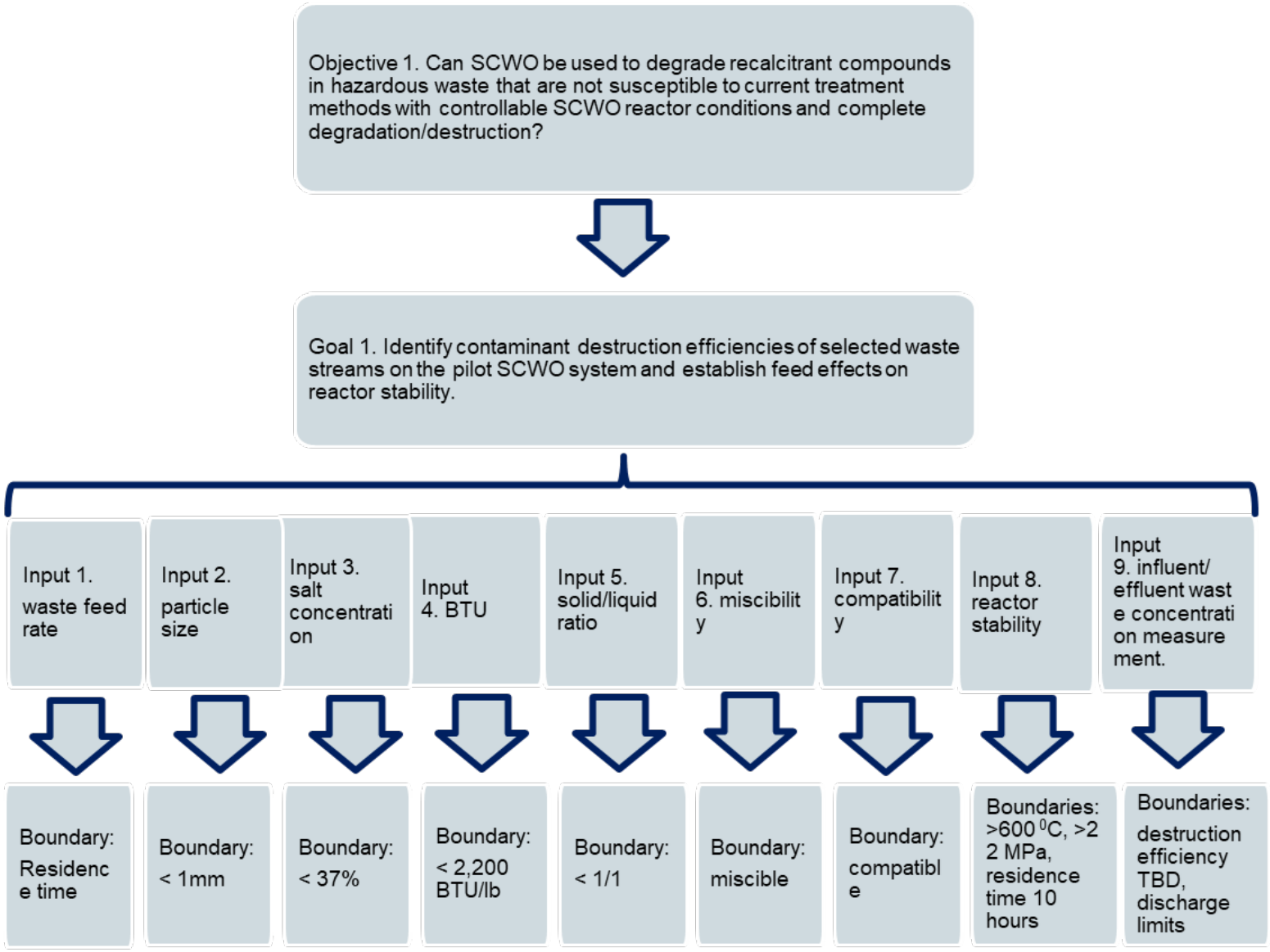

Chart 1: Goals, inputs and boundary conditions to address the DQO 1.

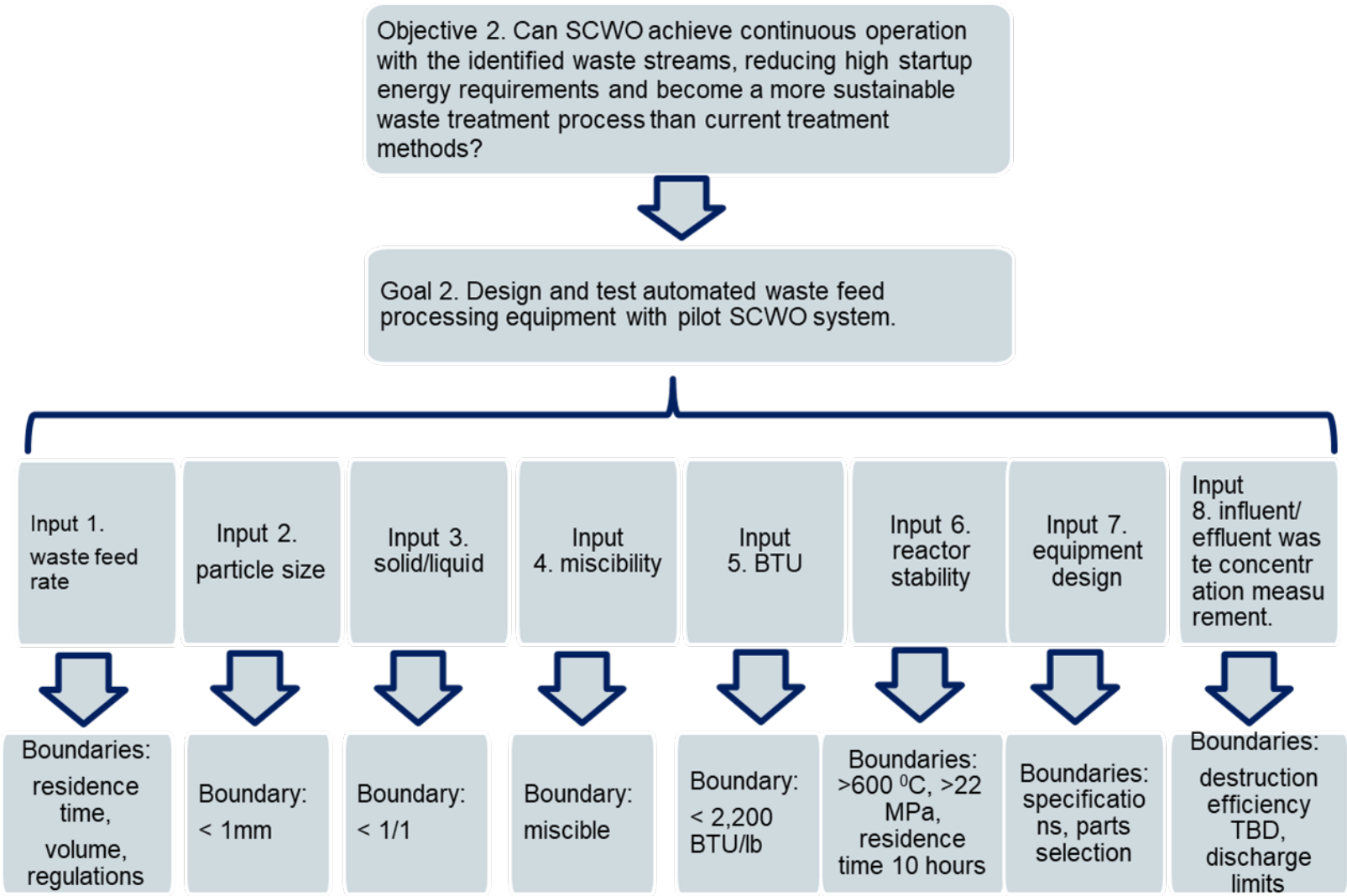

Chart 2: Goals, inputs and boundary conditions to address the DQO 2.

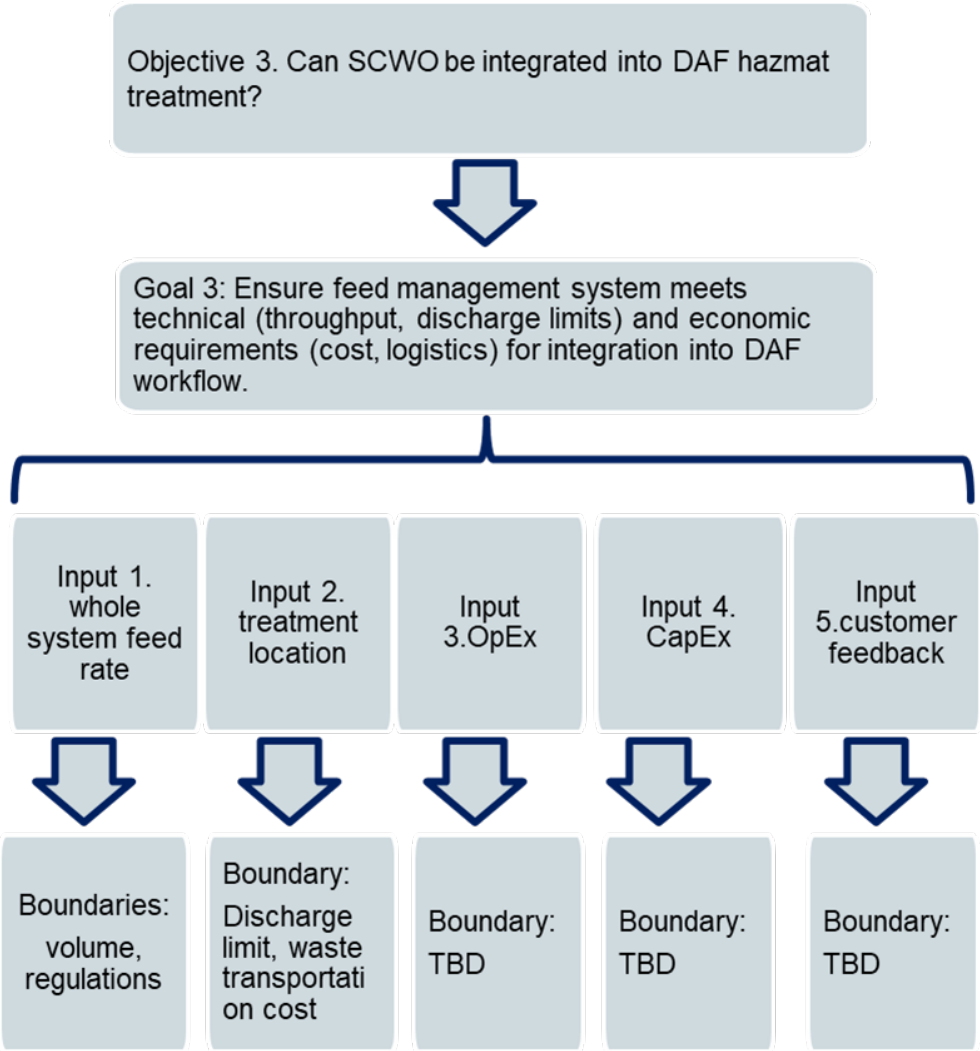

Chart 3: Goals, inputs and boundary conditions to address the DQO 3.

1.6 Next Steps

Upon completion of CLIN001, we will continue to work on:

1. determining wastes that contain DAF stakeholder's key contaminants and for inclusion in our testing.
2. establishing volume requirements and acquire everything needed for waste characterization according to Inputs as listed in above Charts such as waste pumpability, salt content, COD, particle size analysis, etc.
3. arrange for shipping of the waste from the DAF bases to AxNano's pilot SCWO test facility in San Diego for research use only using DOT permitted transporters. Necessary chain of custody documentation, permitting, safe handling guidelines, storage and disposal requirements will be prepared and provided to the transporters and the pilot test facility. If waste cannot be received from bases, we will source from private waste handlers or create synthetic waste.

The next deliverable, CLIN002, is due Nov 21, 2023 and is selection and logistics for wastes to be used in the study. Then CLIN003 is the Sampling Design and Rationale, which is due Jan 21, 2023.

Task	Due Date	Deliverable	Acceptance criteria
Task 1. Establish Systematic Planning Process Statements.	Aug. 21, 2023	List of Data Quality Objectives (DQOs) and Testing Process.	TPOC confirms DQOs and process will inform AFCEC decision makers.
Task 2. Establish logistics for wastes.	Nov. 21, 2023	Report on waste handling and logistics.	TPOC agrees with conclusions drawn in report.
Task 3. Finalize Sampling Design and Rationale.	Jan. 21, 2024	Sampling Design and Rationale (SDR) Report.	TPOC approves SDR.
Task 4. Prepare for test.	Mar. 21, 2024	Test plan, materials and equipment list.	TPOC approves Test Plan.
Task 5. Acquire Materials/Equipment for Pretest.	May 21, 2024	Progress report.	TPOC confirms reasonable effort to Task 5.
Task 6. Run test.	Aug. 21, 2024	Photolog of test.	TPOC confirms test has occurred.
Task 7. Prepare Test Report.	Nov. 21, 2024	Test report.	TPOC agrees with conclusions drawn in report.
Task 8. Survey DAF stakeholders for system integration.	Jan. 21, 2025	Progress report.	TPOC confirms reasonable effort to Task 8.
Task 9. Research required permitting.	Feb. 21, 2025	Permitting report.	TPOC agrees with conclusions drawn in report.
Task 10. Strategize for Phase III.	Apr. 21, 2025	Progress report.	TPOC confirms reasonable effort to Task 10.